Supplementary Material

Systematic benchmarking of functional module partitioning
in GO pathway subnetworks

Supplementary Tables S1--S5

Supplementary Figures S1--S2

Total evaluation records: 2170 (main) + 528 (synthetic) + 317 (downstream)

Table S1. Cohort characteristics and pathway coverage

Accession	Disease	Median pathway genes	Samples	Pathways	Algorithm evaluations
GSE10072	Lung adenocarcinoma	164	107	7	7
GSE137342	Severe sepsis	160	57	7	7
GSE15653	Type 2 diabetes / obesity	117	18	6	6
GSE17536	Colorectal cancer	125	177	7	7
GSE19188	Non-small cell lung cancer	137	156	7	7
GSE25066	Breast cancer	115	486	7	7
GSE36961	Hypertrophic cardiomyopathy	172	145	7	7
GSE39582	Colorectal cancer	102	560	6	6
GSE42568	Breast cancer	86	121	6	6
GSE61635	Systemic lupus erythematosus	31	129	4	4
GSE6764	Hepatocellular carcinoma	101	72	6	6

Table S2. Complete algorithm performance metrics

Algorithm	Modules	Modularity Q	Coherence z	Stability	Robustness	Antagonistic	n
CPM γ=0.20	136.543	0.034	4.581	0.927	0.877	0.992	70
CPM γ=0.10	113.329	0.097	2.689	0.885	0.928	0.962	70
Clique Percolation	42.500	0.047	2.284	0.919	0.920	0.258	70
CPM γ=0.05	91.543	0.151	2.106	0.874	0.947	0.796	70
K-means	5.500	0.143	1.611	0.739	0.882	0.563	70
Ward	5.500	0.146	1.597	0.690	0.852	0.580	70
Gaussian Mixture	5.500	0.140	1.355	0.660	0.793	0.451	70
OPTICS	11.986	0.163	1.210	0.301	0.634	0.700	70
HDBSCAN	4.414	0.199	1.123	0.387	0.758	0.326	70
Louvain γ=0.5	16.743	0.252	1.016	0.501	0.836	0.546	70
DBSCAN	3.800	0.093	1.000	0.517	0.847	0.126	70
Infomap	6.186	0.251	0.988	0.751	0.960	0.369	70
Walktrap	9.714	0.266	0.983	0.577	0.805	0.522	70
Leiden γ=1.0	4.629	0.304	0.972	0.652	0.913	0.459	70
Leiden γ=2.0	26.871	0.212	0.939	0.645	0.890	0.801	70
Louvain	4.686	0.303	0.923	0.648	0.915	0.475	70
Fast Greedy	4.586	0.299	0.897	0.671	0.945	0.467	70
Multilevel	4.643	0.303	0.892	0.650	0.865	0.528	70
Louvain γ=2.0	26.200	0.213	0.890	0.622	0.847	0.849	70
Affinity Propagation	18.786	0.239	0.840	0.682	0.846	0.812	70
Leiden γ=10.0	85.429	0.042	0.840	0.645	0.902	0.989	70
Label Propagation	3.129	0.226	0.795	0.671	0.898	0.134	70
Leiden γ=0.5	2.457	0.264	0.786	0.674	0.918	0.191	70
NMF	5.471	0.188	0.741	0.474	0.775	0.640	70
Leading Eigenvector	4.000	0.279	0.719	0.593	0.905	0.415	70
Leiden γ=5.0	64.700	0.093	0.710	0.649	0.894	0.968	70
Spectral	5.500	0.204	0.703	0.607	0.896	0.474	70
Mean Shift	6.800	0.026	0.392	0.550	0.885	0.110	70
Leiden γ=0.1	1.171	0.133	0.002	0.910	0.955	0.013	70
WGCNA-style Hierarchical	3.000	0.006	-0.101	0.627	0.874	0.008	70
Spin Glass	1.000	—	-0.161	1.000	1.000	0.000	70

Table S3. Paired statistical tests

Metric	Pairs	Median (exp)	Median (ref)	p	q (FDR)	Cliff δ	Sig.
模块数	70	4.500	3.000	8.40e-12	2.10e-11	+0.857	yes
模块度Q	70	0.304	0.000	3.56e-13	1.78e-12	+0.989	yes
功能内聚z	70	0.768	-0.075	3.19e-07	5.32e-07	+0.522	yes
稳定性	70	0.642	0.635	4.95e-01	4.95e-01	+0.064	no
拮抗拆分率	31	0.500	0.000	7.59e-05	9.49e-05	+0.633	yes

Experimental: Leiden γ=1.0. Reference: WGCNA-style hierarchical. Wilcoxon signed-rank, Benjamini-Hochberg correction.

Table S4. Synthetic benchmark results (ground truth available)

Algorithm	LFR mean ARI	LFR max ARI	Nested sub-group ARI	Modules
CPM γ=0.10	0.379	1.000	1.000	12.0
Infomap	0.000	0.000	1.000	12.0
Label Propagation	0.237	1.000	1.000	12.0
Leiden γ=2.0	0.603	0.992	1.000	12.0
Leiden γ=5.0	0.122	0.220	1.000	12.0
Leiden γ=1.0	0.927	1.000	0.943	11.3
Louvain	0.941	1.000	0.943	11.3
Walktrap	0.774	1.000	0.943	11.3
WGCNA-style Hierarchical	0.929	1.000	0.908	11.0
CPM γ=0.05	0.000	0.000	0.863	10.3
Leiden γ=0.5	0.000	0.000	0.602	6.0
Spectral	0.876	1.000	0.602	6.0
Ward	0.724	1.000	0.551	6.0
K-means	0.641	1.000	0.377	6.0
Gaussian Mixture	0.598	1.000	0.352	6.0

LFR: block model across fuzziness $\mu \in \{0.10 – 0.70\}$. Nested: recovery of 12 ground-truth sub-groups (the decisive benchmark).

Table S5. Null-model comparison

Cohort	Pathway	Observed Q	Null A (rewired)	Null B (permuted)	z (A)	z (B)
GSE137342	apoptotic process (GO:0006915)	0.173	0.170	0.419	0.62	-12.42
GSE137342	cellular response to oxidative stress (G	0.165	0.215	0.510	-7.21	-11.96
GSE137342	chemokine-mediated signaling pathway (GO	0.315	0.544	0.583	-4.39	-5.59
GSE137342	inflammatory response (GO:0006954)	0.295	0.226	0.456	10.09	-6.25
GSE137342	regulation of I-kappaB kinase/NF-kappaB	0.171	0.147	0.417	6.52	-10.18
GSE137342	regulation of immune response (GO:005077	0.304	0.280	0.460	1.95	-6.56
GSE137342	response to cytokine (GO:0034097)	0.165	0.205	0.470	-6.22	-13.20
GSE25066	apoptotic process (GO:0006915)	0.366	0.785	0.482	-12.74	-3.94
GSE25066	cellular response to oxidative stress (G	0.213	0.704	0.537	-6.31	-8.46
GSE25066	chemokine-mediated signaling pathway (GO	0.344	0.648	0.566	-3.32	-5.02
GSE25066	inflammatory response (GO:0006954)	0.294	0.644	0.432	-15.51	-6.82
GSE25066	regulation of I-kappaB kinase/NF-kappaB	0.268	0.727	0.485	-14.55	-8.37
GSE25066	regulation of immune response (GO:005077	0.308	0.501	0.474	-7.44	-6.39
GSE25066	response to cytokine (GO:0034097)	0.485	0.717	0.517	-7.48	-0.90
GSE39582	apoptotic process (GO:0006915)	0.544	0.706	0.436	-6.03	4.70
GSE39582	cellular response to oxidative stress (G	0.500	0.730	0.510	-5.86	-0.34
GSE39582	chemokine-mediated signaling pathway (GO	0.529	0.621	0.561	-1.44	-0.71
GSE39582	inflammatory response (GO:0006954)	0.182	0.453	0.437	-15.07	-8.41
GSE39582	regulation of I-kappaB kinase/NF-kappaB	0.490	0.637	0.436	-7.21	2.59
GSE39582	regulation of immune response (GO:005077	0.263	0.394	0.474	-6.26	-7.27
GSE39582	response to cytokine (GO:0034097)	0.483	0.674	0.476	-6.80	0.26
GSE6764	apoptotic process (GO:0006915)	0.286	0.448	0.428	-8.59	-7.51
GSE6764	cellular response to oxidative stress (G	0.324	0.542	0.477	-6.70	-6.44
GSE6764	chemokine-mediated signaling pathway (GO	0.470	0.624	0.558	-2.96	-1.60
GSE6764	regulation of I-kappaB kinase/NF-kappaB	0.389	0.474	0.419	-5.51	-1.66
GSE6764	regulation of immune response (GO:005077	0.404	0.439	0.479	-1.72	-3.00
GSE6764	regulation of inflammatory response (GO:	0.335	0.459	0.459	-11.54	-4.71
GSE6764	response to cytokine (GO:0034097)	0.412	0.522	0.466	-3.45	-2.43

Mean z = -5.54 (A) and -5.09 (B). Negative z indicates that observed modularity does not exceed the null models.

Supplementary Figures

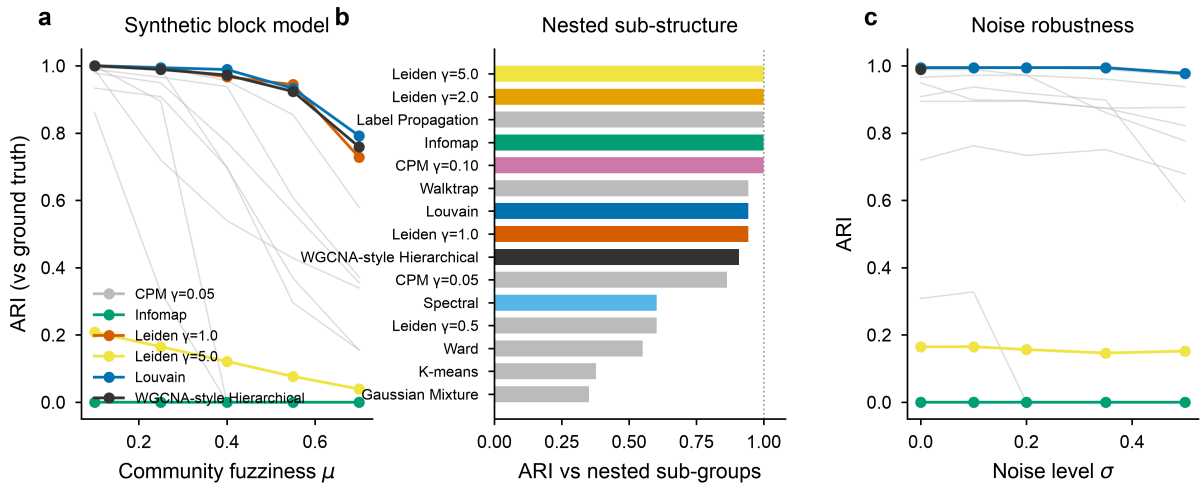


Figure S1. Synthetic benchmark accuracy across fuzziness and noise levels.

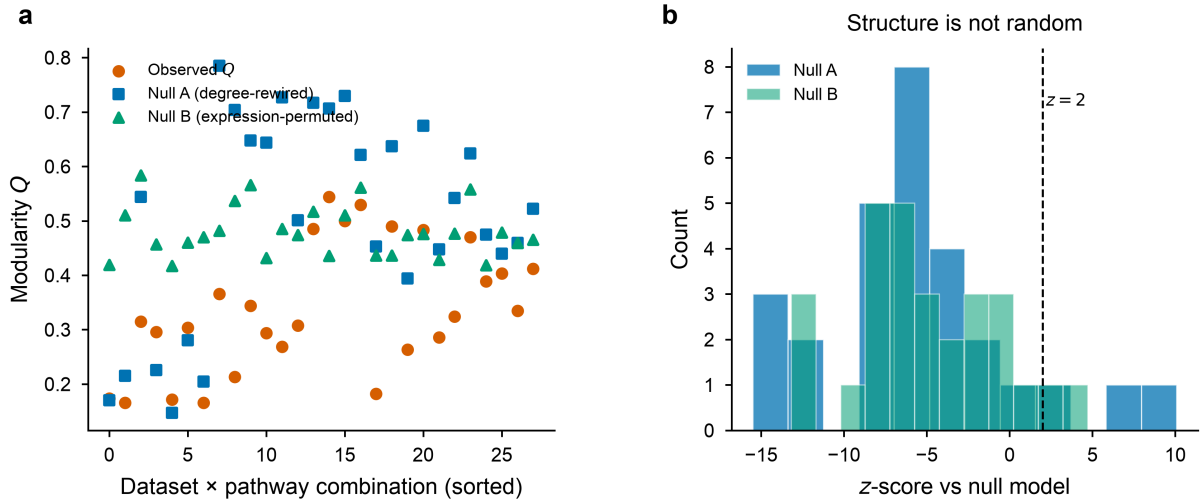


Figure S2. Observed modularity versus null models.